Jianfeng Zhang

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Education

- **Wuhan University** — Wuhan, China
  Bachelor of Science in Applied Mathematics — *Sep. 2014 – Jun. 2018*

Research Interests

- 2D/3D computer vision, deep learning and their applications, including human pose/shape estimation, stereo matching and interactive image segmentation.

Publications

- Xuecheng Nie, **Jianfeng Zhang**, Shuicheng Yan, Jiashi Feng. *Single-stage Multi-person Pose Machines* , ICCV, 2019.
- **Jianfeng Zhang**, Yan Zhu, Xiaoping Zhang, Ming Ye, Jinzhong Yang. *Developing a Long Short-Term Memory (LSTM) based Model for Predicting Water Table Depth in Agricultural Areas*, Journal of Hydrology, 2018.
- **Jianfeng Zhang**, Xuecheng Nie, Hongsong Wang, Pan Zhou, Jiashi Feng. *Nested Graph Auto-Encoder for 3D Human Pose Estimation*, Arxiv, 2019.
- **Jianfeng Zhang**, Liezhuo Zhang, Yuankai Teng, Xiaoping Zhang, Song Wang, Lili Ju. *Interactive Binary Image Segmentation with Edge Preservation*, Arxiv, 2018.

Projects

- **Nested Graph Auto-Encoder for 3D Human Pose Estimation** — LV Lab, Singapore
  Research Assistant — *Feb. 2019 – Jun. 2019*
  - We proposed a novel model to learn 3D structural representation of human skeleton under a weakly-supervised learning strategy. This paper is submitted to NIPS 2019.
- **Semantic Segmentation in RGBD Images** — OrionStar CV Lab, China
  Research Intern — *Oct. 2018 – Jan. 2019*
  - We developed several algorithms to perform image semantic segmentation in RGBD Images in order to help robots to better localize objects position.
- **Interactive Segmentation on RGBD Image** — Farsee2, China
  Research Intern — *Mar. 2018 – Jun. 2018*
  - We replicate ”Interactive Segmentation on RGBD Images via Cue Selection”, this paper proposes a novel interactive segmentation algorithm which can incorporate multiple feature cues like color, depth and normal in a graph cut framework.
- **Video Object Segmentation Framework** — Wuhan University, China
  Research Assistant — *Feb. 2018 – Mar. 2018*
  - We built a video object segmentation framework. In this framework, we used MobileNet to extract each frame’s feature and utilized a simple algorithm to determine if a certain frame is key frame. Based on this method, we could extract expected number of key frames from a given video.
  - We utilized YOLOv2 and Deep Grabcut to detect center object and obtain object binary mask, respectively.
  - Finally, we used key frame masks to finetune segmentation network before segment the whole video. Only 200-3000 iters of online finetune could ensure very good video object segmentation results.
  - This framework is currently applied to extract products in E-commerce videos and people in videos from Douyin, Kuaishou and YouTube.
- **Clothing Fashion Style Recognition** — *Jun. 2018 – Jul. 2018*
  - We built fashion style recognition model based on ResNet, DenseNet and VGGnet. We also used Deeplab V3+ to extract object of interest and remove background, which could help to denoise. In addition, we applied model ensemble method to improve our results. Finally, our method achieved 0.610 F2 score and ranked **9/213** on JD AI Fashion-Challenge.

Work Experience

- **Research Assistant at NUS LV Lab** *Feb. 2019 – Present*
- **R&D intern at OrionStar Inc.** *Oct. 2018 – Jan. 2019*
- **R&D intern at Farsee2 Inc.** *Oct. 2017 – Oct. 2018*

Awards & Honors

- **Excellent Demo Award, DeeCamp** *2018*
- **Best Bachelor Thesis Award** *2018*
- **Award for Merit Student of Wuhan University** *2015-2018*
- **Honorable Mention, MCM, COMAP** *2015*

Programming Skills

- **Languages: Python, C/C++, C#, Java**
- **Deep Learning Tools: Tensorflow, PyTorch**